

Cheeger-type inequalities for the second largest spectral gap from 1 of the normalized Laplacian

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based on joint work with Lies Beers and Raffaella Mulas

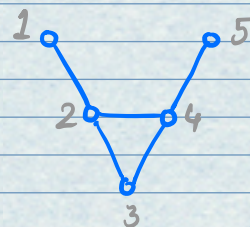
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Normalized Laplacian of a graph

$G = (V, E)$ with no isolated vertices, $|V| = n$
adjacency matrix A
degree matrix D



$$A = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix} \quad D = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

(left) normalized Laplacian $L = Id - D^{-1}A$

eigenvalues $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$

$$L = \begin{pmatrix} 1 & -1 & 0 & 0 & 0 \\ -1/3 & 1 & -1/3 & -1/3 & 0 \\ 0 & -1/2 & 1 & -1/2 & 0 \\ 0 & -1/3 & -1/3 & 1 & -1/3 \\ 0 & 0 & 0 & -1 & 1 \end{pmatrix}$$

The eigenvalues of L satisfy:

$$0 \leq (7 - \sqrt{13})/6 \leq 1 \leq 5/3 \leq (7 + \sqrt{13})/6$$

- multiplicity of 0 is the number of connected components of G
(in particular, $\lambda_1 = 0$)

$$\frac{n}{n-1} \leq \lambda_n \leq 2$$

equality iff $G = K_n$

equality iff at least one component is bipartite

Two applications of L :

- DL can be used to compute the number of spanning trees of G (Kirchhoff, 1847)
- bounding mixing times of random walks

Rayleigh quotients

$C(V)$... (real) vector space of functions $f: V \rightarrow \mathbb{R}$

inner product $\langle \cdot, \cdot \rangle: C(V) \times C(V) \rightarrow \mathbb{R}$

$$\langle f, g \rangle = \sum_{v \in V} \deg(v) \cdot f(v) \cdot g(v)$$

L is self-adjoint with respect to $\langle \cdot, \cdot \rangle$, i.e. $\langle Lf, g \rangle = \langle f, Lg \rangle$

Rayleigh quotient of $g \in C(V) \setminus \{0\}$: $RQ_L(g) = \frac{\langle Lg, g \rangle}{\langle g, g \rangle}$

Thm (Courant - Fischer - Weyl min-max principle):

Let $B \in \mathbb{R}^{V \times V}$ be self-adjoint with respect to $\langle \cdot, \cdot \rangle$,
with eigenvalues $\beta_1 \leq \dots \leq \beta_m$.

Then, for all $i \in [n]$:

$$\beta_i = \min_{\substack{H \subseteq C(V) \\ \dim H = i}} \max_{\substack{g \in H \\ g \neq 0}} RQ_L(g) = \max_{\substack{H \subseteq C(V) \\ \dim H = n-i+1}} \min_{\substack{g \in H \\ g \neq 0}} RQ_L(g).$$

Second largest spectral gap from 1

$$\tau = \max_{i \neq 1} |1 - \lambda_i| = \max \{1 - \lambda_2, \lambda_n - 1\}$$

$\tau = 1$ iff G is disconnected or bipartite
Why study τ ?

- bounds the rate of convergence to the stationary distribution of the (lazy) simple random walk on G
- Ramanujan graphs: connected d -regular graphs such that

$$\max_{i: \lambda_i \neq 0, 2} |1 - \lambda_i| \leq \frac{2\sqrt{d-1}}{d}$$

(non-bipartite Ramanujan graphs satisfy $\tau \leq \frac{2\sqrt{d-1}}{d}$)

Set $M = (Id - L)^2 = (D^{-1}A)^2$. Its eigenvalues are $(1 - \lambda_1)^2, \dots, (1 - \lambda_n)^2$. $M_{vu} = \sum_{w \in \mathcal{N}(v) \cap \mathcal{N}(u)} \frac{1}{\deg w \cdot \deg v} \geq 0$.

$$RQ_M(f) = \frac{1}{\sum_{w \in V} \deg w \cdot f(w)^2} \sum_{w \in V} \frac{1}{\deg w} \cdot \left(\sum_{v \in \mathcal{N}(w)} f(v) \right)^2$$

Related work: analysing $\varepsilon = \min_{i \neq 1} |1 - \lambda_i|$ (Jost, Mulas, Zhang, 2023)

Cheeger constants and inequalities

For $S \subseteq V$, define $\bar{S} = V \setminus S$

$$\text{vol } S = \sum_{w \in S} \deg w$$

and if $S \neq \emptyset$,
$$h(S) = \frac{e(S, \bar{S})}{\text{vol } S}$$

The Cheeger constant of G is
$$h = \min_{\substack{\emptyset \neq S \subseteq V \\ \text{vol } S \leq \text{vol } V/2}} h(S).$$

(Pólya, Szegő, 1951)

The dual Cheeger constant of G is
$$\bar{h} = \max_{\substack{V = S_1 \dot{\cup} S_2 \dot{\cup} S_3 \\ S_1, S_2 \neq \emptyset}} \frac{2 \cdot e(S_1, S_2)}{\text{vol}(S_1) + \text{vol}(S_2)}$$

(Bauer, Hua, Jost, 2014)

Cheeger inequalities:
$$1 - \sqrt{1 - h^2} \leq \lambda_2 \leq 2h$$

Dual Cheeger inequalities:
$$2\bar{h} \leq \lambda_n \leq 1 + \sqrt{1 - (1 - \bar{h})^2}.$$

Bounding τ using Cheeger and dual Cheeger constants

Set $\hat{h} = \min \{h, 1-\bar{h}\}$. Recall $\tau = \max \{1-\lambda_2, \lambda_n-1\}$,
so $1-\tau = \min \{\lambda_2, 2-\lambda_n\}$.

Cheeger inequalities:

$$1 - \sqrt{1-h^2} \leq \lambda_2 \leq 2h$$

Dual Cheeger inequalities: $1 - \sqrt{1-(1-\bar{h})^2} \leq 2-\lambda_n \leq 2(1-\bar{h})$

Since $f(x) = 1 - \sqrt{1-x^2}$ is increasing on $[0, 1]$,
we have

$$1 - \sqrt{1-\hat{h}^2} \leq 1-\tau \leq 2\hat{h}.$$

Here, we use a minimum of two constants.

Could there be a single "natural" constant
for bounding τ ?

Cheeger-type constant $\tilde{h}$

For $\emptyset \neq S \subseteq V$, define $\tilde{h}(S) = \frac{1}{\text{vol } S} \sum_{w \in V} \frac{e(w, S)^2}{\deg w}$

and set $\tilde{h} = \max_{\substack{\emptyset \neq S \subseteq V \\ \text{vol } S \leq \text{vol } V/2}} \tilde{h}(S)$.

Probabilistic interpretation: $\tilde{h}(S) = \sum_{w \in V} \frac{e(w, S)}{\text{vol } S} \cdot \frac{e(w, S)}{\deg w}$

Assume that we select a starting vertex with probability proportional to its degree. Then $\tilde{h}(S)$ is the probability that a random walk starting in S returns to S after two steps.

Compare with $h(S) = \frac{e(S, \bar{S})}{\text{vol } S}$, the probability that a one-step random walk starting in S leaves S .

Proposition: $\tilde{h} \leq 1$, equality iff G is bipartite or disconnected.

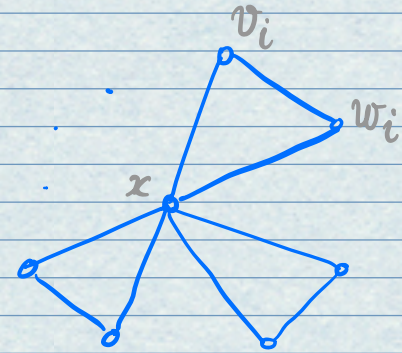
Proof idea: $\tilde{h}(S) = \frac{1}{\sum_{w \in S} \deg w} \cdot \sum_{w \in V} \frac{1}{\deg w} \cdot e(w, S)^2 = \frac{\sum_{w \in V} \frac{1}{\deg w} \left(\sum_{v \in V} \mathbb{1}_S(v) \right)^2}{\sum_{w \in V} \deg w \cdot \mathbb{1}_S(w)^2}$
 $= RQ_M(\mathbb{1}_S)$

Example: m -petal graphs

Let $\emptyset \neq S \subseteq V$ s.t. $\text{vol } S \leq \text{vol } V/2$.

Let $c = \mathbb{1}_{\{x \in S\}}$,

$$k = \sum_{\substack{v \in V \\ v \neq x}} \mathbb{1}_{\{v \in S\}}.$$



Then $\text{vol } S = 2mc + 2k$. Since $\text{vol } S \leq \text{vol } V/2 = 3m$,

either $c=0$ & $k \leq \lfloor \frac{3m}{2} \rfloor$, $\Rightarrow \tilde{h}(S) = \frac{k+m}{4m}$

or $c=1$ & $k \leq \lfloor \frac{m}{2} \rfloor$. $\Rightarrow \tilde{h}(S) = \frac{k+2m}{4m}$

$$\tilde{h}(S) = \frac{1}{\text{vol } S} \cdot \sum_{w \in V} \frac{e(w, S)^2}{\deg w} = \frac{1}{2mc+2k} \cdot \left[\frac{k^2}{2m} + k \cdot \frac{(1+c)^2}{2} + (2m-k) \cdot \frac{c^2}{2} \right]$$

In both cases, $\tilde{h}(S)$ is increasing in k , and hence

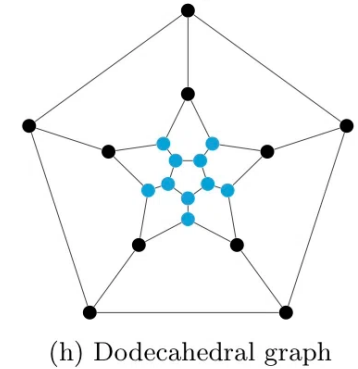
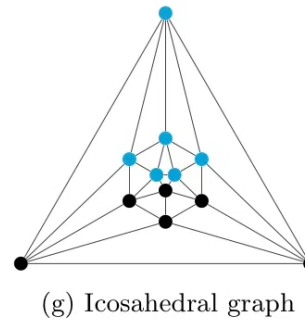
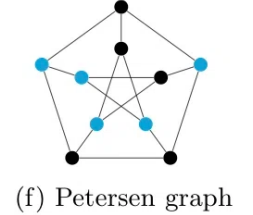
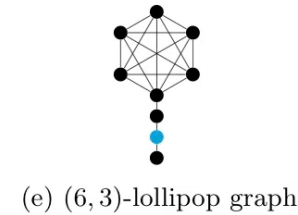
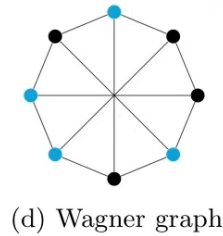
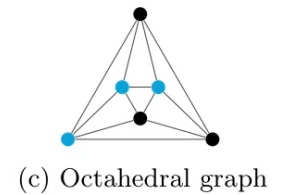
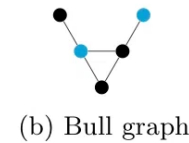
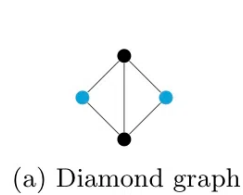
$$\tilde{h} = \begin{cases} \frac{5}{8} & \text{if } m \text{ is even} \\ \frac{5m-1}{8m} & \text{if } m \text{ is odd} \end{cases}$$

More examples

• K_n : $\tilde{h} = \begin{cases} \frac{1}{2} \left(1 + \frac{1}{(n-1)^2} \right) & \text{if } n \text{ is even} \\ \frac{1}{2} \left(1 - \frac{n-3}{(n-1)^2} \right) & \text{if } n \text{ is odd} \end{cases}$

• C_{2m+1} : $\tilde{h} = \frac{2m-1}{2m}$

Graph	n	$\tilde{h}$
Diamond graph	4	2/3
Bull graph	5	17/24
Octahedral graph	6	7/12
Wagner graph	8	7/9
(6,3)-lollipop graph	9	3/4
Petersen graph	10	31/45
Icosahedral graph	12	3/5
Dodecahedral graph	20	7/9



Question: Is there a function f s.t. $\tilde{h} \geq f(n)$ for all graphs on n vertices, satisfying $\lim_{n \rightarrow \infty} f(n) = \frac{1}{2}$?

Main theorem - Cheeger-type inequality

Thm (BMP, 26+): $\sqrt{1 - (1 - \tilde{h})^2} \geq \tau^2 \geq 2\tilde{h} - 1$

[Compare with $1 - \sqrt{1 - h^2} \leq \lambda_2 \leq 2h$
 $2\bar{h} \leq \lambda_n \leq 1 + \sqrt{1 - (1 - \bar{h})^2}$
 $1 - 2\hat{h} \leq \tau \leq \sqrt{1 - \hat{h}^2}.]$

Proof of the lower bound $\tau^2 \geq 2\tilde{h} - 1$

Thm (BMP, 26+): $\tau^2 \geq \max_{\emptyset \neq S \subseteq V} \underbrace{\left(\frac{\text{vol } V}{\text{vol } S} \cdot \left(\frac{1}{\text{vol } S} \sum_{w \in V} \frac{e(w, S)^2}{\deg w} \right) - \frac{\text{vol } S}{\text{vol } S} \right)}_{(*)}$

Thm $\Rightarrow \tau^2 \geq 2\tilde{h} - 1$:

Let $\emptyset \neq S \subseteq V$ such that $\text{vol } S \leq \text{vol } V/2$.

By Thm,
$$\begin{aligned} \tau^2 &\geq \frac{\text{vol } V}{\text{vol } S} \cdot \tilde{h}(S) - \frac{\text{vol } S}{\text{vol } S} \\ &= \tilde{h}(S) + \underbrace{\frac{\text{vol } S}{\text{vol } S}}_{\leq 1} \underbrace{(\tilde{h}(S) - 1)}_{\leq 0} \\ &\geq 2\tilde{h}(S) - 1 \end{aligned}$$

□

Proof sketch of Thm: Fix $\emptyset \neq S \subseteq V$.

Define $f: V \rightarrow \mathbb{R}$ by $f(v) = \begin{cases} 1, & v \in S \\ -\frac{\text{vol } S}{\text{vol } S}, & v \in \bar{S} \end{cases}$.

Then $\tau^2 \geq RQ(f) = \dots = (*)$

Note: for bipartite and disconnected graphs $\tau^2 = 1 = 2\tilde{h} - 1$.

Proof of the upper bound $\tau^2 \leq \sqrt{1 - (1 - \tilde{h})^2}$

Recall $\tau^2 = \max_{f \in C(V) \setminus \{0\}} RQ_M(f) = \max_{f \in C(V) \setminus \{0\}} \frac{\sum_{w \in V} \frac{1}{\deg w} \left(\sum_{v \in N(w)} f(v) \right)^2}{\sum_{w \in V} \deg w \cdot f(w)^2}$

$\sum_{w \in V} \deg w \cdot f(w) = 0$ $\sum_{w \in V} \deg w \cdot f(w) = 0$

Step 1: Find $f \in C(V) \setminus \{0\}$ s.t.

- $\tau^2 \leq RQ_M(f)$
 - $\text{vol supp}(f) \in (0, \frac{\text{vol } V}{2}]$
 - $1 \in \text{Range}(f) \subset [0, 1]$
- } useful for a probabilistic argument

Step 2: Find $\emptyset \neq S \subset V$ satisfying

- $RQ_M(f) \leq \sqrt{1 - (1 - \tilde{h}(S))^2}$
- $\text{vol } S \leq \text{vol } V / 2$

Proof of the upper bound $\tau^2 \leq \sqrt{1 - (1 - \tilde{h})^2}$

Let $g \in C(V) \setminus \{0\}$ satisfy

$$\sum_{w \in V} \deg w \cdot g(w) = 0 \quad \text{and} \quad \tau^2 = RQ_M(g)$$

$\Rightarrow g$ contains at least one positive and one negative value

For any $c \in \mathbb{R}$ we have $RQ_M(g - c \cdot 1) \geq RQ_M(g)$

Pick c such that $\text{vol} \{v : g(v) - c \geq 0\} \geq \frac{\text{vol } V}{2}$
and $\text{vol} \{v : g(v) - c \leq 0\} \geq \frac{\text{vol } V}{2}$

In particular, $\text{vol} \{v : g(v) - c < 0\} \leq \frac{\text{vol } V}{2}$
and $\text{vol} \{v : g(v) - c > 0\} \leq \frac{\text{vol } V}{2}$

Set $\tilde{g} = g - c \cdot 1$, $\tilde{g}^+ = \max\{\tilde{g}, 0\}$, $\tilde{g}^- = \max\{-\tilde{g}, 0\}$, so $\tilde{g} = \tilde{g}^+ - \tilde{g}^-$

$$\text{vol supp}(\tilde{g}^+) \leq \frac{\text{vol } V}{2}, \quad \text{vol supp}(\tilde{g}^-) \leq \frac{\text{vol } V}{2}$$

Step 1: Find $f \in C(V) \setminus \{0\}$ s.t.

- $\tau^2 \leq RQ_M(f)$
- $\text{vol supp}(f) \in (0, \frac{\text{vol } V}{2}]$
- $1 \in \text{Range}(f) \subset [0, 1]$

Step 2: Find $\emptyset \neq S \subset V$ satisfying

- $RQ_M(f) \leq \sqrt{1 - (1 - \tilde{h}(S))^2}$
- $\text{vol } S \leq \text{vol } V/2$

Proof of the upper bound $\tau^2 \leq \sqrt{1 - (1 - \tilde{h})^2}$

Claim: At least one of $\tilde{f} \in \{\tilde{g}^+, \tilde{g}^-\}$ satisfies $\tilde{f} \neq 0$ and $RQ_M(\tilde{g}) \leq RQ_M(\tilde{f})$.

Step 1: Find $f \in C(V) \setminus \{0\}$ s.t.

- $\tau^2 \leq RQ_M(f)$
- $\text{vol supp}(f) \in (0, \frac{\text{vol } V}{2}]$
- $1 \in \text{Range}(f) \subset [0, 1]$

Pf of Claim: At least one of $\{\tilde{g}^+, \tilde{g}^-\}$ is non-zero.

Step 2: Find $\emptyset \neq S \subset V$ satisfying

- $RQ_M(f) \leq \sqrt{1 - (1 - \tilde{h}(S))^2}$
- $\text{vol } S \leq \text{vol } V/2$

If $\tilde{g}^+ = 0$, set $\tilde{f} = \tilde{g}^-$ (and vice versa)

If both are non-zero:

$$RQ_M(\tilde{g}) = \frac{\langle \tilde{g}^+ - \tilde{g}^-, M(\tilde{g}^+ - \tilde{g}^-) \rangle}{\langle \tilde{g}^+ - \tilde{g}^-, \tilde{g}^+ - \tilde{g}^- \rangle}$$

$$= \frac{\langle \tilde{g}^+, M\tilde{g}^+ \rangle - \langle \tilde{g}^+, M\tilde{g}^- \rangle - \langle \tilde{g}^-, M\tilde{g}^+ \rangle + \langle \tilde{g}^-, M\tilde{g}^- \rangle}{\langle \tilde{g}^+, \tilde{g}^+ \rangle + \langle \tilde{g}^-, \tilde{g}^- \rangle}$$

$$\leq \frac{\langle \tilde{g}^+, M\tilde{g}^+ \rangle + \langle \tilde{g}^-, M\tilde{g}^- \rangle}{\langle \tilde{g}^+, \tilde{g}^+ \rangle + \langle \tilde{g}^-, \tilde{g}^- \rangle}$$

$$\leq \max \{ RQ_M(\tilde{g}^+), RQ_M(\tilde{g}^-) \}$$

□

Finally, set $f = \frac{1}{\max \{ \tilde{f}(v) : v \in V \}} \cdot \tilde{f}$, concluding Step 1.

Proof of the upper bound $\tau^2 \leq \sqrt{1 - (1 - \tilde{h})^2}$

Consider the random set $S_t = \{v : f(v) \geq t\}$,
where $t^2 \sim U[0, 1]$.

Then: $\bullet IP[v \in S_t] = IP[f(v)^2 \geq t^2] = f(v)^2$

$$\bullet IP[u, v \in S_t] = \min\{f(u)^2, f(v)^2\}$$

$$\bullet IP[u \in S_t \vee v \in S_t] = \max\{f(u)^2, f(v)^2\}$$

$$\bullet IP[S_t = \emptyset] = 0 \quad (\text{since } 1 \in \text{Range}(f))$$

$$\bullet S_t \subset \text{supp}(f) \text{ almost surely } (IP[t=0]=0)$$

$$\Rightarrow \text{vol } S_t \in (0, \text{vol } V/2) \text{ almost surely}$$

$$\bullet E[\text{vol } S_t] = \sum_{w \in V} \deg w \cdot E[\mathbb{1}_{S_t}(w)] = \sum_{w \in V} \deg w \cdot f(w)^2$$

$$RQ_M(f) = \frac{1}{\sum_{w \in V} \deg w \cdot f(w)^2} \sum_{w \in V} \frac{1}{\deg w} \cdot \left(\sum_{v \in N(w)} f(v) \right)^2$$

$$= \frac{1}{\sum_{w \in V} \deg w \cdot f(w)^2} \sum_{w \in V} \frac{1}{\deg w} \sum_{(u,v) \in N(w)^2} f(u) \cdot f(v)$$

Step 1: Find $f \in C(V) \setminus \{0\}$ s.t.

- $\tau^2 \leq RQ_M(f)$
- $\text{vol } \text{supp}(f) \in (0, \frac{\text{vol } V}{2}]$
- $1 \in \text{Range}(f) \subset [0, 1]$

Step 2: Find $\emptyset \neq S \subset V$ satisfying

- $RQ_M(f) \leq \sqrt{1 - (1 - \tilde{h}(S))^2}$
- $\text{vol } S \leq \text{vol } V/2$

Proof of the upper bound $\tau^2 \leq \sqrt{1 - (1 - \tilde{h})^2}$

$$RQ_M(f) = \frac{1}{\sum_{w \in V} \deg w \cdot f(w)^2} \sum_{w \in V} \frac{1}{\deg w} \cdot \left(\sum_{v \in N(w)} f(v) \right)^2$$

$$= \frac{1}{\sum_{w \in V} \deg w \cdot f(w)^2} \sum_{w \in V} \frac{1}{\deg w} \sum_{(u,v) \in N(w)^2} \underbrace{f(u)f(v)}_{\min\{f(u), f(v)\} \cdot \max\{f(u), f(v)\}}$$

Step 1: Find $f \in C(V) \setminus \{0\}$ s.t.

- $\tau^2 \leq RQ_M(f)$
- $\text{vol supp}(f) \in (0, \frac{\text{vol } V}{2}]$
- $1 \in \text{Range}(f) \subset [0, 1]$

Step 2: Find $\emptyset \neq S \subset V$ satisfying

- $RQ_M(f) \leq \sqrt{1 - (1 - \tilde{h}(S))^2}$
- $\text{vol } S \leq \text{vol } V/2$

C-S

$$\leq \frac{1}{\sum_{w \in V} \deg w \cdot f(w)^2} \left(\sum_{\substack{w \in V \\ (u,v) \in N(w)^2}} \frac{\min^2}{\deg w} \right)^{1/2} \left(\sum_{\substack{w \in V \\ (u,v) \in N(w)^2}} \frac{\max^2}{\deg w} \right)^{1/2}$$

$$= \frac{1}{\mathbb{E}[\text{vol } S_t]} \left(\sum \frac{\mathbb{P}[u,v \in S_t]}{\deg w} \right)^{1/2} \cdot \left(\sum \frac{\mathbb{P}[u \in S_t \vee v \in S_t]}{\deg w} \right)^{1/2}$$

$$= \frac{1}{\mathbb{E}[\text{vol } S_t]} \left(\mathbb{E} \sum \frac{\mathbb{1}\{u,v \in S_t\}}{\deg w} \right)^{1/2} \left(\mathbb{E} \sum \frac{\mathbb{1}\{u \in S_t \vee v \in S_t\}}{\deg w} \right)^{1/2}$$

Proof of the upper bound $\tau^2 \leq \sqrt{1 - (1 - \tilde{h})^2}$

Note: $\sum_{(u,v) \in E(w)^2} \mathbb{1}\{u,v \in S_t\} = e(w, S_t)^2$

$$\begin{aligned} \sum_{(u,v) \in E(w)^2} \mathbb{1}\{u \in S_t \vee v \in S_t\} &= \\ \sum_{(u,v) \in E(w)^2} \mathbb{1}\{u,v \in S_t\} + \sum_{(u,v) \in E(w)^2} \mathbb{1}\{u \in S_t \vee v \in S_t\} &= \\ \frac{e(w, S_t)^2}{\deg w} + 2e(w, S_t)e(w, \bar{S}_t) \end{aligned}$$

$$RQ_M(f) \leq \frac{1}{\mathbb{E}[\text{vol } S_t]} \left(\mathbb{E} \sum_{\substack{w \in V \\ (u,v) \in E(w)^2}} \frac{\mathbb{1}\{u,v \in S_t\}}{\deg w} \right)^{1/2} \left(\mathbb{E} \sum_{\substack{w \in V \\ (u,v) \in E(w)^2}} \frac{\mathbb{1}\{u \in S_t \vee v \in S_t\}}{\deg w} \right)^{1/2}$$

$$= \frac{1}{\mathbb{E}[\text{vol } S_t]} \cdot \left(\mathbb{E} \left[\sum_{w \in V} \frac{e(w, S_t)^2}{\deg w} \right] \right)^{1/2} \cdot \left(\mathbb{E} \left[\sum_{w \in V} \frac{e(w, S_t)^2}{\deg w} + \frac{2e(w, S_t) \cdot e(w, \bar{S}_t)}{\deg w} \right] \right)^{1/2}$$

$$\begin{aligned} &= \dots \\ &= \frac{\mathbb{E}[X_t]^{1/2} \cdot (-\mathbb{E}[X_t] + 2\mathbb{E}[Y_t])^{1/2}}{\mathbb{E}[Y_t]} \end{aligned}$$

$$= \sqrt{1 - \left(1 - \frac{\mathbb{E}[X_t]}{\mathbb{E}[Y_t]}\right)^2}$$

Step 1: Find $f \in C(V) \setminus \{0\}$ s.t.

- $\tau^2 \leq RQ_M(f)$
- $\text{vol supp}(f) \in (0, \frac{\text{vol } V}{2}]$
- $1 \in \text{Range}(f) \subset [0, 1]$

Step 2: Find $\emptyset \neq S \subset V$ satisfying

- $RQ_M(f) \leq \sqrt{1 - (1 - \tilde{h}(S))^2}$
- $\text{vol } S \leq \text{vol } V/2$

$$X_t := \sum_{w \in V} \frac{e(w, S_t)^2}{\deg w}$$

$$Y_t := \text{vol } S_t$$

(note: $\mathbb{P}[Y_t > 0] = 1$)

Proof of the upper bound $\tau^2 \leq \sqrt{1 - (1 - \tilde{h})^2}$

$$RQ_M(f) \leq \sqrt{1 - \left(1 - \frac{\mathbb{E}[X_t]}{\mathbb{E}[Y_t]}\right)^2}$$

Lemma: Let X and Y be random variables such that $P[Y > 0] = 1$.
Then $P\left[\frac{X}{Y} \geq \frac{\mathbb{E}[X]}{\mathbb{E}[Y]}\right] > 0$.

$$RQ_M(f) \leq \sqrt{1 - \left(1 - \frac{X_{t_0}}{Y_{t_0}}\right)^2} \quad \text{for some } t_0.$$

Using that $f(x) = \sqrt{1 - (1 - x)^2}$ is increasing on $[0, 1]$:

$$RQ_M(f) \leq \sqrt{1 - (1 - \tilde{h}(S_0))^2}$$

Concluding step 2 and the whole proof. $\square$

Step 1: Find $f \in C(V) \setminus \{0\}$ s.t.

- $\tau^2 \leq RQ_M(f)$
- $\text{vol supp}(f) \in (0, \frac{\text{vol } V}{2}]$
- $1 \in \text{Range}(f) \subset [0, 1]$

Step 2: Find $\emptyset \neq S \subset V$ satisfying

- $RQ_M(f) \leq \sqrt{1 - (1 - \tilde{h}(S))^2}$
- $\text{vol } S \leq \text{vol } V / 2$

$$X_t := \sum_{w \in V} \frac{e(w, S_t)}{\deg w}$$

$$Y_t := \text{vol } S_t$$

(note: $P[Y_t > 0] = 1$)

Proof of the upper bound $\tau^2 \leq \sqrt{1 - (1 - \tilde{h})^2}$

Equality case: G is disconnected or bipartite:

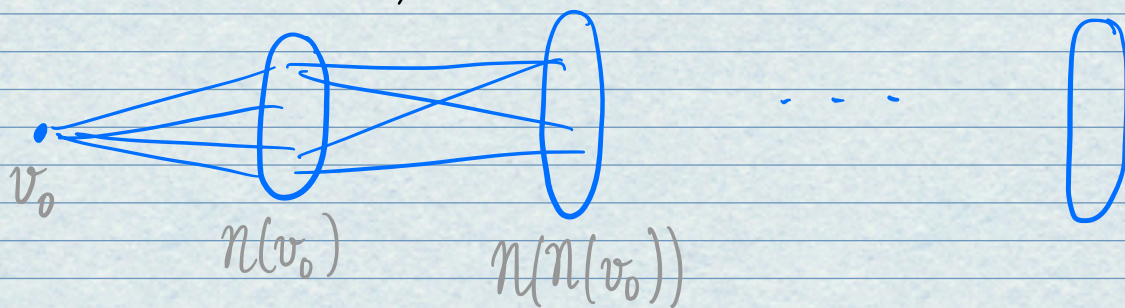
Needed $\sum_{\substack{w \in V \\ (u,v) \in N(w)^2}} \frac{\min\{f(u), f(v)\} \cdot \max\{f(u), f(v)\}}{\deg w} = (\text{Cauchy-Schwarz})$

$$\left(\sum_{\substack{w \in V \\ (u,v) \in N(w)^2}} \frac{\min^2}{\deg w} \right)^{1/2} \cdot \left(\sum_{\substack{w \in V \\ (u,v) \in N(w)^2}} \frac{\max^2}{\deg w} \right)^{1/2}$$

This implies $f(u) = f(v)$ for all $w \in V$ and all $(u,v) \in N(w)^2$.

If G is disconnected, then $\tilde{h} = 1$, $\tau = 1$.

If G is connected:



$S_0 (S_1) \dots$ vertices at an even (odd) distance from v_0

f is constant on S_0 and S_1 , and attains at least two values by construction $\Rightarrow S_0$ and S_1 are independent sets

$\Rightarrow G$ is bipartite. □

Proof of the upper bound $\tau^2 \leq \sqrt{1 - (1 - \tilde{h})^2}$

Remark: By applying Bernoulli's inequality, one gets

$$\tau^2 \leq 1 - \frac{(1 - \tilde{h})^2}{2}$$

with equality iff G is disconnected or bipartite

THANK YOU FOR YOUR ATTENTION!